

Lecture 9

31/08/24 10:05 AM

$$X = \frac{Y-b}{a}$$

$$cdf = \int pdf$$

- Functions of continuous r.v.s:

$$F_Y(y) = P(Y \leq y)$$

$$= P(aX+b \leq y)$$

$$= P\left(X \leq \frac{y-b}{a}\right) = F_X\left(\frac{y-b}{a}\right)$$

Consider $Y = aX + b$

↳ continuous r.v.

$$F_Y(y) = P(Y \leq y) = P(aX+b \leq y) = \begin{matrix} a > 0 & a < 0 \\ P\left(X \leq \frac{y-b}{a}\right) & P\left(X > \frac{y-b}{a}\right) \end{matrix}$$

$$= F_X\left(\frac{y-b}{a}\right), \text{ if } a > 0 \quad / \quad \bar{F}_X\left(\frac{y-b}{a}\right), \text{ if } a < 0$$

$$\text{If } a > 0: f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} F_X\left(\frac{y-b}{a}\right)$$

$$= \frac{d}{du} F_X(u) \cdot \frac{du}{dy} = \frac{1}{a} \cdot f_X(u)$$

$$= \begin{matrix} \frac{1}{a} \cdot f_X\left(\frac{y-b}{a}\right) & / & \frac{-1}{a} \cdot f_X\left(\frac{y-b}{a}\right) \\ a > 0 & & a < 0 \end{matrix}$$

*** If $Y = g(X)$ is continuous, differentiable and monotone:

Let $h(\cdot)$ denote the inverse fn. of $g(\cdot)$

$$\Rightarrow h(Y) = X$$

$$\text{Then } f_Y(y) = \left| \frac{d}{dy} h(y) \right| \cdot f_X(h(y))$$

REFER SLIDES !!!

- MIXED RANDOM VARIABLES

neither continuous nor discrete

- MULTIPLE RANDOM VARIABLES

$$\text{joint pmf: } p_{XY}(x,y) = P(\{\omega \in \Omega: X(\omega)=x \text{ and } Y(\omega)=y\})$$

$$\text{joint cdf: } F_{XY}(x,y) = P(\{\omega \in \Omega: X(\omega) \leq x \text{ and } Y(\omega) \leq y\})$$